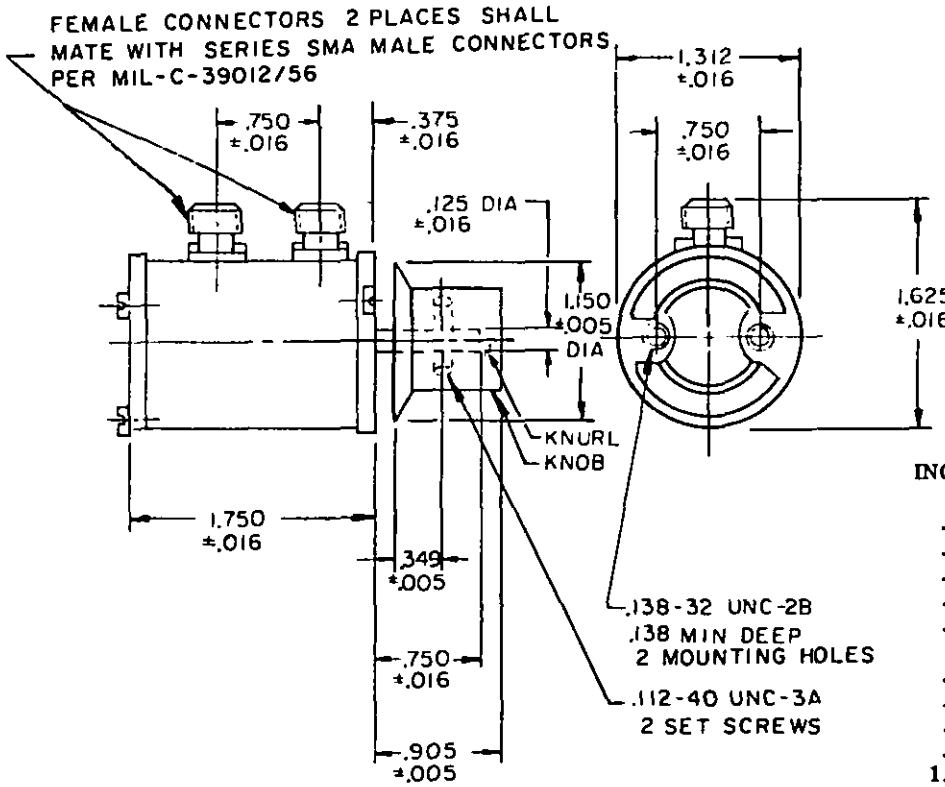
MILITARY SPECIFICATION SHEET

ATTENUATORS, VARIABLE (COAXIAL),
FREQUENCY RANGE DC TO 2 GHZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the attenuators described herein shall consist of this document and the latest issue of Specification MIL-A-24215.



INCHES	MM
.005	.13
.016	.41
.112	2.84
.125	3.18
.138	3.51
.349	8.86
.375	9.53
.750	19.05
.905	22.99
1.150	29.21
1.312	33.32
1.625	41.28
1.750	44.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Dimensions and configuration.

FSC 5985

Source: https://assist.dla.mil -- Downloaded: 20260-19T20:00Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Operating frequency: See table I.

Insertion loss: See table I.

Input and output impedance: 50 ohms.

Voltage standing wave ratio (VSWR): See table I.

Attenuation range: See table I.

Attenuation accuracy: See table I.

Rotation angle between steps: 30°.

Input maximum average power (derate linearly to zero watts at 125 C): 0.5 watts @ 25°C.

Operating ambient temperature: -20 C to +125 C.

Final finish: All exposed surfaces and unexposed aluminum surfaces shall be chemically treated per MIL-C-5541 type 1, grade B, class 2.

Weight: 5 ounces maximum.

Calibrated knob: Rogan Bros. Inc. part number RB-67-1-SK, or equivalent.

Part number: See table I.

TABLE I.

Part number	Attenuation range (dB)	Accuracy	Insertion loss (max)	VSWR (max)	Frequency range
M24215/8-001	0 to 10 dB in 1 dB steps	±0.1 dB at .5 GHz ±0.3 dB at 1 GHz ±0.5 dB at 2 GHz	0.5 dB	1.15:1 at .5 GHz 1.15:1 at 1 GHz 1.25:1 at 2 GHz	DC to 2 GHz
-002	0 to 1 dB in 0.1 dB steps	±0.03 dB at .5 GHz ±0.05 dB at 1 GHz	1.0 dB	1.10:1 at .5 GHz 1.20:1 at 1 GHz	DC to 1 GHz

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:00Z
Check the source to verify that this is the current version before use.

NOTE: Attenuators covered by this specification sheet are substitutable for the following *commercial types:*

Part number	FSN 5985	Manufacturer's code	Commercial type
M24215/8-001		49956	2887497-1
-001	493-5749	80249	331029-1
-001	493-5749	23042	MA-50
-002		49956	2887497-2
-002		23042	MA-53

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI, SL, MU
Navy -
Air Force - 11
DSA - ES

User activities:
Army - SM
Navy - MC, AS, OS, SH
Air Force -

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-T160)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:00Z
Check the source to verify that this is the current version before use.